

Lecture 1

Question 1: What is the primary goal of a recommendation system?

- A. Classification
- B. Clustering
- C. Prediction
- D. Personalized suggestion

Question 2: In collaborative filtering, what does the "user-item matrix" represent?

- A. Users' demographic information
- B. Items' characteristics
- C. Users' preferences for items
- D. Ratings given by users to items

Question 3: Which evaluation metric measures the accuracy of predicted ratings in recommendation systems?

- A. Precision
- B. Recall
- C. Mean Absolute Error (MAE)
- D. F1 score

Question 4: In the context of collaborative filtering, what is the purpose of the user-based approach?

- A. To identify similar items
- B. To find similar users
- C. To recommend popular items
- D. To address the cold start problem

Question 5: Which algorithm is commonly used for generating recommendations in an item-item collaborative filtering system?

- A. k-Nearest Neighbors (k-NN)
- B. Decision Trees
- C. Apriori algorithm
- D. Naive Bayes

Lecture 2

Question 1: What is the primary objective of clustering algorithms in data mining?

- A. Classification
- B. Prediction
- C. Grouping similar data points
- D. Outlier detection

Question 2: What is the main drawback of the k-means clustering algorithm?

- A. Sensitivity to initial cluster centroids
- B. Inability to handle non-linear clusters
- C. Lack of scalability
- D. Dependency on data distribution

Question 3: How does the DBSCAN algorithm handle outliers in the dataset?

- A. Assigns outliers to the nearest cluster
- B. Removes outliers from the dataset
- C. Labels outliers as noise
- D. Forms separate clusters for outliers

Question 4: Which clustering algorithm is particularly effective for identifying clusters with varying shapes and densities?

- A. K-means
- B. DBSCAN
- C. Hierarchical clustering
- D. None of above

Question 5: Given the 1D data points {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, if k-means clustering is run with $k=2$ and initial cluster centers at 2 and 8, how many points will be in the cluster with center 8 in the first iteration?

- A. 4
- B. 5
- C. 6
- D. 7

Lecture 3

Question 1: What does covariance measure in the context of data mining?

- A. Strength of linear relationship between two variables
- B. Independence between two variables
- C. Magnitude of individual variables
- D. Mean absolute difference between two variables

Question 2: In eigenvalue decomposition of a matrix, what is the significance of the eigenvectors?

- A. They represent the scaling factors
- B. They represent the rotation angles
- C. They define the transformation matrix
- D. They define the principal components

Question 3: In PCA, what is the primary goal of transforming the original variables into principal components?

- A. To reduce the dimensionality of the data.
- B. To increase the interpretability of the data.
- C. To increase the correlation between variables.
- D. To make the data non-linear.

Question 4: In the context of SVD, what is the significance of singular values?

- A. They represent the magnitude of eigenvalues
- B. They indicate the strength of covariance between variables
- C. They define the principal components
- D. They quantify the importance of individual dimensions

Question 5: What is a key advantage of SVD over eigenvalue decomposition?

- A. SVD is faster
- B. SVD is more accurate
- C. SVD works only for square matrices
- D. SVD can be applied to non-square matrices

Lecture 4

Question 1: In a self-organizing map (SOM), which neuron's weights are adjusted during training?

- A. Only the winning neuron
- B. The winning neuron and its neighbors
- C. All neurons in the map
- D. A randomly selected subset of neurons

Question 2: Given a trained SOM and a new input vector $[0.2, 0.7]$, which of the following neurons would be selected as the winner using Euclidean distance?

- A. Neuron 1: $[0.1, 0.5]$
- B. Neuron 2: $[0.3, 0.8]$
- C. Neuron 3: $[0.6, 0.4]$
- D. Neuron 4: $[0.8, 0.2]$

Question 3: Multidimensional scaling (MDS) aims to ensure that points close together in the embedded space are also close in the original space. What term quantifies the reconstruction error?

- A. Accuracy
- B. Stress
- C. Entropy
- D. Inertia: the distance between each data point and its centroid

Question 4: What is the main advantage of using t-Distributed Stochastic Neighbor Embedding (t-SNE) over traditional methods like Principal Component Analysis (PCA)?

- A. Faster computation
- B. Better preservation of local structures in the data
- C. Reduced risk of overfitting
- D. Improved interpretability

Lecture 5

Question 1: What is the primary purpose of encoding in the context of searching and ranking in data mining?

- A. Compression of data
- B. Conversion of data into a format
- C. Creation of an index for efficient retrieval
- D. Encryption of data

Question 2: In the context of searching and ranking in data mining, what is indexing used for?

- A. Sorting data for presentation
- B. Efficient retrieval of relevant information
- C. Encoding data for storage
- D. Encrypting sensitive information

Question 3: What does the term "inverted index" refer to in the context of searching and ranking?*

- A. An index that is sorted in reverse order
- B. An index where the positions of terms in documents are recorded
- C. A type of encoding technique
- D. A ranking algorithm

Question 4: Which of the following is an essential component of the matching process in searching and ranking?

- A. Clustering
- B. Indexing
- C. Ranking
- D. Similarity measure

Question 5: How does tf-idf (Term Frequency-Inverse Document Frequency) contribute to the ranking of documents in information retrieval?

- A. It measures the frequency of a term in a document
- B. It penalizes common terms and emphasizes those with a high frequency in a given document
- C. It encrypts the document content
- D. It compresses the document for efficient storage

Question 6: How is TF-IDF calculated for a term w in a document d ? f , F are the frequency of w , the frequency of all the words appeared in d . N , n are the total number of documents, and the number of documents where the term w appears.

- A. $F/f * \log (N/n)$
- B. $f/F * \log (N/n)$
- C. $F/f * \log (n/N)$
- D. $f/F * \log (n/N)$

Lecture 6

Question 1: What is the key assumption made by the Naive Bayes classifier?

- A. Features are correlated
- B. Features are independent given the class
- C. Features follow a Gaussian distribution
- D. Features have equal variance

Question 2: What is the purpose of the "assumed" parameter in the Naive Bayes classifier for text data?

- A. It is the assumed value for the prior probability
- B. It is the assumed value for the posterior probability
- C. It is the assumed value for the likelihood probability when a word is not present
- D. It is the assumed value for the feature independence assumption

Question 3: In Fisher's method, the combined p-value is calculated by:

- A. Adding the individual p-values
- B. Multiplying the individual p-values
- C. Taking the average of the individual p-values
- D. Using a chi-square statistic on the individual p-values

Question 4: Suppose we have a binary classification problem with two classes: spam (c_1) and not spam (c_2). The prior probabilities are $P(c_1) = 0.4$ and $P(c_2) = 0.6$. For a given document d , the likelihood probabilities are $P(d|c_1) = 0.7$ and $P(d|c_2) = 0.3$. If we use the Naive Bayes method, is this document d a spam or not?

- A. Yes
- B. No

Lecture 7

Question 1: Which of the following is NOT a characteristic of decision trees?

- A. Tree-like structure with nodes representing feature tests
- B. Branches represent outcomes of feature tests
- C. Leaf nodes represent class labels
- D. Nodes must split data into equal-sized partitions

Question 2: What is the primary goal when constructing a decision tree for classification?

- A. To maximize the depth of the tree
- B. To create a balanced tree
- C. To minimize the number of leaf nodes
- D. To maximize the purity of leaf nodes

Question 3: If a node contains 10 examples, 6 of class A and 4 of class B, what is the Gini impurity of this node?

- A. 0.48
- B. 0.52
- C. 0.64
- D. 0.36

Question 4: In the context of decision trees, what does the term "greedy" refer to?

- A. The algorithm's appetite for more data
- B. The algorithm's tendency to overfit
- C. The algorithm's ability to find the globally optimal solution
- D. The algorithm's strategy of making the locally optimal choice at each step

Question 5: Which ensemble method builds multiple decision trees on random subsets of features?

- A. Bagging
- B. Boosting
- C. Random forests
- D. None of the above

Lecture 8

Question 1: What is the key idea behind the k-NN algorithm?

- A. Use training records directly to predict the class label of test cases by considering their neighbor correlations
- B. Train an explicit model using the training data
- C. Use majority voting to determine the class label
- D. a and c

Question 2: Given the following data points and their class labels: (1, 1, Class A), (2, 2, Class B), (3, 3, Class A), (4, 4, Class B), what would be the class label assigned to the unknown point (2, 3) using the k-NN algorithm with $k=3$ and Euclidean distance?

- A. Class A
- B. Class B
- C. Tie between Class A and Class B
- D. Not enough information to determine the class label

Question 3: What is a potential issue when using the k-NN algorithm with different attribute scales?

- A. The distance measures may be dominated by one of the attributes
- B. The algorithm may not converge
- C. The algorithm may overfit the training data
- D. The algorithm may be biased towards the minority class

Question 4: What is the purpose of using Locality-Sensitive Hashing (LSH) in the context of k-NN?

- A. To improve the accuracy of the algorithm
- B. To reduce the computational complexity of the algorithm
- C. To handle missing data in the dataset
- D. To perform dimensionality reduction

Question 5: Which of the following is NOT a disadvantage of the k-NN algorithm?

- A. It is sensitive to the choice of k
- B. It requires storing all the training data
- C. It is not suitable for large datasets due to computational complexity
- D. It cannot handle non-linear decision boundaries

Lecture 9

Question 1: Which of the following statements best describes a transaction in market basket analysis?

- A. A set of all items sold in a store
- B. An individual item or article in a basket
- C. A set of items purchased together in a single shopping basket
- D. The process of generating association rules

Question 2: What is the purpose of the Apriori algorithm in market basket analysis?

- A. To find all possible association rules
- B. To find all frequent itemsets that satisfy the minimum support threshold
- C. To generate high-confidence rules from frequent itemsets
- D. Both b) and c)

Question 3: The anti-monotone property of support in the Apriori algorithm states that:

- A. If an itemset is frequent, then all of its supersets must also be frequent
- B. If an itemset is infrequent, then all of its subsets must also be infrequent
- C. If an itemset is frequent, then all of its subsets must also be frequent
- D. If an itemset is infrequent, then some of its supersets might be frequent

Question 4: In a dataset with 5000 transactions, the itemset {X, Y} appears in 1200 transactions, and the itemset {X} appears in 2000 transactions. What is the confidence of the association rule $\{X\} \rightarrow \{Y\}$?

- A. 0.24
- B. 0.4
- C. 0.6
- D. 2.4

Lecture 10

Question 1: What is the underlying principle behind Latent Semantic Analysis (LSA)?

- A. Identifying co-occurrence patterns of words in text data
- B. Applying supervised learning on labeled text data
- C. Extracting topics using clustering techniques
- D. Finding semantic features with dimensionality reduction on the term-document matrix

Question 2: In the context of CLIP, what is the purpose of the contrastive learning approach?

- A. To cluster text and image data into distinct groups
- B. To align the representations of semantically associated text-image pairs
- C. To generate textual descriptions for images
- D. To translate between text and image data

Question 3: Which of the following is a limitation of LSA that CLIP aims to address?

- A. Inability to handle negation and logical operations
- B. Difficulty in representing abstract concepts not directly present in the text
- C. Lack of robustness to noise and sparsity in data
- D. Inability to capture cross-modal semantic relationships between text and images

Question 4: Which of the following statements is correct regarding the interpretability of the learned representations in LSA and CLIP?

- A. Both LSA and CLIP produce highly interpretable representations
- B. LSA produces more interpretable representations than CLIP
- C. CLIP produces more interpretable representations than LSA
- D. Neither LSA nor CLIP produces interpretable representations

Lecture 11

Question 1: Topic modeling can be conceptualized as:

- A. A supervised classification method for text data
- B. A process of uncovering the underlying themes or topics in a collection of documents
- C. A method for extracting keywords from documents
- D. A dimensionality reduction technique for text data

Question 2: What is the main advantage of probabilistic topic models like PLSA and LDA over deterministic models like NMF?

- A. They can handle uncertainties and complexities in textual data more effectively
- B. They are more computationally efficient
- C. They can automatically determine the number of topics
- D. They produce more interpretable topics

Question 3: In the context of Latent Dirichlet Allocation (LDA), what is the role of the Dirichlet prior distribution?

- A. It is used to model the distribution of words given a topic
- B. It is used to model the distribution of topics given a document
- C. It is used to model the distribution of documents given a corpus
- D. Both (a) and (b)

Question 4: Which of the following inference techniques is commonly used for parameter estimation in PLSA?

- A. Expectation-Maximization (EM) algorithm
- B. random sampling
- C. k-means clustering
- D. Both (a) and (b)

Lecture 12

Question 1: What is the main idea behind using a Gaussian Mixture Model (GMM) for outlier detection?

- A. Data points with low probability density under the fitted GMM are considered outliers
- B.) Data points that belong to low-weight Gaussian components are marked as outliers
- C. Data points far away from the mean of any Gaussian component are outliers
- D. Both a) and c)

Question 2: Which of the following is NOT a parameter that needs to be learned when fitting a GMM?

- A. Mean of each Gaussian component
- B. Covariance matrix of each Gaussian component
- C. Weight/Mixing proportion of each Gaussian component
- D. Number of Gaussian components in the mixture

Question 3: Which of the following is a potential advantage of using DBSCAN over Gaussian mixture models for outlier detection?

- A. DBSCAN can handle data with varying densities better
- B. DBSCAN does not require setting any parameters like neighborhood radius
- C. DBSCAN is more computationally efficient for high-dimensional data
- D. DBSCAN can automatically determine the optimal number of clusters

Question 4: Which statement best describes a key difference between GMMs and DBSCAN for outlier detection?

- A. GMMs are probabilistic models, while DBSCAN relies on density estimation
- B. GMMs can only identify outliers, while DBSCAN can find clusters and outliers
- C. GMMs assume Gaussian distributions, while DBSCAN makes no distribution assumptions
- D. GMMs can handle outliers automatically, while DBSCAN requires pruning outliers